

■ FARMSKING PLATFORM

Full Developer Architecture, Technical Specification & Production
Deployment Guide

Author: Core Engineering Team & Lead Architect

System Version: v1.0 Production Ready

Platform: React Native (Expo) + NestJS Backend + PostgreSQL Prisma + Sentinel-2 Satellite GIS

■ PURPOSE OF THIS DOCUMENTATION

This comprehensive developer blueprint is designed for technical teams, system administrators, and stakeholders. It breaks down the complete software architecture of **FarmsKing**, explaining how all frontend mobile modules, backend microservices, database schemas, and external satellite GIS integrations interact seamlessly. It also outlines key market differentiators, legal compliance policies (Google Play, IT Act 2000, GDPR), pre-launch audit checklists, and long-term product roadmaps.

Chapter 1: Executive Overview & Technical Blueprint

1.1 System Architecture Overview

FarmsKing is a full-stack agricultural management ecosystem built with modern, battle-tested technologies. The architecture is designed to handle low-connectivity rural mobile usage, real-time double-entry ledgers, and high-resolution satellite imagery processing.

Architecture Layer	Technology Stack	Role & Description
Frontend Mobile	React Native, Expo SDK 52, Expo Router v4, TypeScript	Cross-platform mobile application for Android & Web with offline caching & dynamic UI theme switching.
Backend API	Node.js, NestJS Framework, TypeScript, REST & WebSockets	Modular backend service handling business logic, authentication, GIS calculations, and background jobs.
Database & ORM	PostgreSQL, Prisma ORM	Relational database with strict FK constraints, ACID transactions, and spatial indexing.
Caching & Storage	Redis, AWS S3 / Google Cloud Storage (GCS)	High-speed session/OTP caching and secure media storage for receipts & crop photos.
GIS & Satellites	Sentinel-2, ISRO Earth Observation Data, OpenLayers / Leaflet	High-resolution optical satellite imagery, 4-corner polygon area math, and NDVI vegetation index generation.
Real-Time Voice	WebSockets, WebRTC, Expo Audio Engine	Live multi-party group voice calls between farmers and agronomist advisors.

1.2 Multi-Role System Architecture

FarmsKing operates on a unified multi-role framework. A single mobile installation dynamically renders customized user interfaces based on the authenticated user's assigned role:

- **Farmer Mode:** Farm plot management, crop activity tracking, Mandi Arhtiya loan ledger, daily worker management, crop disease reporting, and voice AI mic.
- **Advisor Mode:** Multi-farmer roster, 1-click GIS satellite field inspection, pending crop technical audits, disease diagnosis & prescription scheduling, and group voice calls.
- **Arhtiya / Trader Mode:** Mandi crop sale entries, advance loan records with auto monthly compound interest, digital J-Form issuance, and net balance statement exports.
- **Labour / Worker Mode:** Worker attendance login, work log verification, wage calculation, and payment receipt tracking.
- **Super Admin Portal:** System configuration, coupon management, global user audits, backup snapshot downloads, and system health telemetry.

Chapter 2: Feature Matrix & Implementation Workflows

■ ■ 4-Corner Satellite Field GPS Locking & NDVI Diagnostics

Farmers or advisors walk the farm or select remote coordinates to mark 4 corners (C1-NorthWest, C2-NorthEast, C3-SouthEast, C4-SouthWest). The app computes the exact polygon area in Acres using Earth radius trigonometric formulas. Once saved, the location is locked for ISRO/Sentinel-2 optical satellite scans, rendering live NDVI vegetation health scores (0.0 to 1.0) and soil moisture percentages.

■ ■ Mandi Arhtiya Ledger & Auto-Interest Calculation Engine

Eliminates handwritten diaries for Mandi traders and farmers. Supports advance loan tracking with configurable monthly interest rates (e.g. 1.5%/month). Automatically converts mandi weight units (Quintal, 50kg Bori, 35kg Bori, 40kg Mann) into metric Quintals and calculates gross crop sale values. Supports Digital J-Form records and dynamic double-entry net settlement calculation.

■ Labour Worker Management & Attendance Khata

Allows farmers to register wage workers, tractor drivers, and harvest laborers. Tracks work entries (Sowing, Harvesting, Spraying, Weeding) multiplied by daily/hourly rates. Maintains a complete double-entry Khata statement with payment logs and 1-tap WhatsApp statement sharing.

■ ■ ■ Dual Farmer-Advisor Real-Time Collaboration & WebRTC Voice

Farmers submit crops for agronomist review. Advisors perform remote satellite inspection, approve/reject crop cycles, and prescribe spray schedules. Includes a WebRTC-powered Live Group Voice Call system allowing advisors to conduct broadcast voice consultations with multiple farmers.

■ ■ AI Voice Assistant Mic (Voice Command Parser)

A floating mic button converts spoken voice commands into structured financial entries (e.g., 'Bought 2 bags DAP for 2700 rupees') using AI NLP parsing.

■ Digital JPG Invoice & Payment Receipt Exporter

Generates pixel-perfect digital receipts for sales and payments using `react-native-view-shot`, allowing 1-click sharing as high-quality JPG images via WhatsApp.

Chapter 3: Database Schema & Entity Relationships

3.1 Relational Architecture & Table Linkages

The PostgreSQL schema managed by Prisma ORM utilizes strict foreign keys and index constraints. Below is the complete architectural mapping of the core database tables:

Table Name	Primary / Foreign Keys	Key Fields & Attributes	Linked Relationships
User	id (PK) kingId (UNIQUE)	name, mobile, email, role, language, isActive	1:1 FarmerProfile 1:1 AdvisorProfile 1:N Crops
FarmerProfile	id (PK) userId (FK)	sprayTankSizeL, soilType, waterType, village, district	Belongs to User
Farm / Plot	id (PK) ownerId (FK)	name, areaAcres, latitude, longitude	1:N CropCycle
CropCycle	id (PK) plotId (FK) advisorId (FK)	cropName, variety, sowingDate, status, isApproved	1:1 CropGpsData 1:N CropProblem
CropGpsData	id (PK) cropCycleId (FK)	centerLat, centerLng, corners (JSON), areaAcres, isLocked	Belongs to CropCycle
UnifiedParty	id (PK) farmerId (FK)	name, mobile, roles (ARHTIYA, SUPPLIER), mandiName	1:N ArhtiyaAdvance 1:N ArhtiyaCropSale
ArhtiyaAdvance	id (PK) partyId (FK)	amount, interestRateMonthly, transactionDate, status	Belongs to UnifiedParty
ArhtiyaCropSale	id (PK) partyId (FK)	cropName, inputUnit, inputQuantity, ratePerQuintal, jFormNumber	Belongs to UnifiedParty
LabourWorker	id (PK) farmerId (FK)	name, mobile, address, defaultRate, defaultUnit	1:N WorkEntry 1:N LabourPayment
LabourWorkEntry	id (PK) workerId (FK)	workDate, workType, quantity, rate, totalAmount	Belongs to LabourWorker
LabourPayment	id (PK) workerId (FK)	paymentDate, amount, paymentMode, notes	Belongs to LabourWorker

Chapter 4: Market USPs & Rare Features

4.1 Competitive Advantage Analysis

FarmsKing introduces groundbreaking innovations that address real operational challenges faced by rural Indian farmers and Mandi traders:

Feature Benchmark	Traditional Agri Apps (Agrostar, DeHaat)	FarmsKing Advanced Platform
Mandi Finance & Ledger	■ Basic cash register or missing entirely.	■ Integrated Arhtiya Advance & Mandi Sale Ledger with auto 1.5% monthly compound interest & Digital J-Form.
GIS Field Boundary	■ Single-point location (Lat/Lng center only).	■ 4-Corner Polygon GPS Lock (NW, NE, SE, SW) with trigonometric acreage calculation & satellite NDVI sync.
Labour Management	■ None. Managed manually on paper.	■ Built-in Labour Khata: Attendance, Work Log (Sowing/Harvesting), and Payment Ledger with PDF export.
Role Ecosystem	■ Separate apps or single farmer view.	■ 5 Roles in 1 Single App: Farmer, Advisor, Mandi Trader, Worker, Super Admin.
Receipt Distribution	■ In-app view only or PDF requirement.	■ 1-Click JPG Image Export via ViewShot for instant WhatsApp sharing with non-tech users.

Chapter 5: Legal & Regulatory Compliance Framework

5.1 Google Play Store Policy Requirements

- **Delete Account Mandate:** Google Play requires an in-app Account Deletion flow and an accessible web deletion page for users to request permanent removal of their personal profile & ledger data.
- **Privacy Policy Accessibility:** A publicly accessible HTTPS Privacy Policy link must be declared in Google Play Console and rendered inside the application.
- **Runtime Permissions:** Fine/Coarse Location (GPS), Camera (Crop Disease Photos), and Storage permissions must be dynamically requested with clear user explanation rationale.
- **Financial Ledger Disclaimer:** App ledgers must explicitly state that digital entries serve as record-keeping assistance and do not replace statutory banking instruments.

5.2 Information Technology Act, 2000 (India Cyber Law)

- **Section 43A (Data Protection):** Mandates reasonable security practices and procedures for sensitive personal data (mobile numbers, financial transactions, GPS coordinates).
- **Section 79 (Intermediary Guidelines):** FarmsKing acts as an intermediary for farmer-advisor communications and Mandi records, requiring prompt takedown mechanisms for illegal content.
- **Data Audit Logs:** Maintenance of immutable audit logs (`AuditLog` entity) capturing IP address, timestamp, and modification details for financial transactions.

5.3 International Data Standards (GDPR & ISO 27001)

- **Right to Erasure & Portability:** Farmers can export complete transaction records in JSON/CSV format and request hard purges of stored images.
- **ISO/IEC 27001 Security Standards:** Mandatory TLS 1.3 encryption in transit and AES-256 encryption at rest for production databases.

Chapter 6: Pre-Launch Audit & Production Readiness

6.1 Mandatory Pre-Launch Checklist

Category	Audit Checkpoint Item	Status & Requirement
Environment & Secrets	Enforce strict <code>.env`</code> isolation. Remove hardcoded JWT secrets, DB passwords, and API keys.	REQUIRED BEFORE LAUNCH
SSL / HTTPS	Configure Let's Encrypt / Cloudflare TLS 1.3 certificate for domain <code>`api.farmsking.com`</code> .	REQUIRED BEFORE LAUNCH
CORS & Rate Limiting	Restrict CORS origins to approved domain. Enable NestJS <code>`ThrottlerModule`</code> (100 req/min).	REQUIRED BEFORE LAUNCH
Database Indexing	Verify indexes on <code>`kingId`</code> , <code>`mobile`</code> , <code>`farmerId`</code> , <code>`partyId`</code> , <code>`cropCycleId`</code> for sub-50ms queries.	REQUIRED BEFORE LAUNCH
Automated Backups	Enable daily automated PostgreSQL snapshots to Google Cloud Storage (GCS) with 30-day retention.	REQUIRED BEFORE LAUNCH
App Bundle (.aab)	Generate production <code>`aab`</code> Android App Bundle via EAS Build (<code>`eas build -p android`</code>).	READY FOR RELEASE

Chapter 7: Growth & Marketing Promotion Strategy

1. Arhtiya (Commission Agent) Network Viral Loop

Target Grain Mandi commission agents first. Once an Arhtiya adopts FarmsKing to manage advances and J-Forms, they onboard all 200–500 associated farmers onto the app using their King ID referral link.

2. WhatsApp Viral Receipt Sharing

Every time a sale, advance, or worker payment receipt is generated, farmers share it directly as a high-quality JPG image on WhatsApp. The image includes a footer: 'Generated via FarmsKing - Download App' creating organic viral acquisition.

3. Kisan Mela & Agri Retailer Partnerships

Partner with local pesticide & fertilizer seed shops. Retailers distribute FarmsKing QR code cards offering 30-day Pro plan access with seed purchases.

Chapter 8: Future Technology Roadmap

Phase	Innovation Feature	Technical Description
Phase 1 (Q3 2026)	AI Pest Diagnosis via Camera Vision	Integrate Gemini / TensorFlow Lite model to analyze leaf photos and diagnose diseases instantly with 95%+ accuracy.
Phase 2 (Q4 2026)	IoT Soil Sensor & Weather Station	Connect Bluetooth / LoRaWAN IoT soil probes to measure NPK values and soil moisture in real-time.
Phase 3 (Q1 2027)	Drone Spray Booking Marketplace	Enable farmers to book agricultural spray drones on-demand based on satellite crop acreage.
Phase 4 (Q2 2027)	ML Mandi Price Forecasting Engine	Time-series machine learning model forecasting crop prices across top 100 Mandis 14 days in advance.

End of Developer Blueprint Document